FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Main)-2016

ANSWERS, HINTS & SOLUTIONS

PART TEST– IV
(MAIN)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**D**	**B**	**A**
2.	**B**	**C**	**C**
3.	**C**	**A**	**B**
4.	**D**	**B**	**C**
5.	**B**	**D**	**B**
6.	**C**	**D**	**A**
7.	**B**	**C**	**C**
8.	**B**	**B**	**B**
9.	**C**	**A**	**A**
10.	**A**	**C**	**A**
11.	**C**	**A**	**B**
12.	**D**	**D**	**C**
13.	**B**	**A**	**A**
14.	**D**	**A**	**A**
15.	**C**	**A**	**D**
16.	**C**	**C**	**A**
17.	**C**	**B**	**A**
18.	**C**	**A**	**A**
19.	**C**	**D**	**A**
20.	**D**	**A**	**D**
21.	**C**	**D**	**D**
22.	**C**	**D**	**D**
23.	**C**	**D**	**D**
24.	**C**	**C**	**B**
25.	**C**	**D**	**B**
26.	**B**	**C**	**C**
27.	**B**	**D**	**A**
28.	**D**	**C**	**B**
29.	**C**	**D**	**B**
30.	**A**	**C**	**C**

HINTS AND SOLUTION

Physics

PART – I

1. $\frac{3}{2v}-\frac{1}{\infty}=\frac{3/2-1}{30}$

$\therefore$ v = + 90 cm

$\frac{4}{3f}-\frac{3}{2\times 90}=\frac{(4/3)-(3/2)}{-60}$

$\therefore$ $f \approx 69$ cm.

2. Refractive index of bubble with respect to water is less than 1. So shifting of the fringe pattern would be towards B.

3. $R = \rho\frac{\ell}{A} = \rho\ \frac{\ell}{\pi d^2/4}$

$R = \frac{4\rho\ell}{\pi d^2}$

$\frac{\Delta R}{R}\times 100 = \left(\frac{\Delta\ell}{\ell}+2\frac{\Delta d}{d}\right)\times 100 = 4.05\ \%$

4. use $ev = \frac{hc}{\lambda}+W$

6. $\frac{1}{V}-\frac{1}{U}=\frac{1}{f}$

$y=\frac{1}{V}$, $x=\frac{1}{U}$, $c=\frac{1}{f}$

$y - x = c$

$y = x + c$

Intercept

c = 0.2 /cm from graph

$f=\frac{1}{c}=5$cm

7. Time at which no. of un-decayed nuclei become 0.37 N_0 is equal to the mean life of the sample.

$t_{av}=\frac{1}{\lambda}$

$7.5 = \frac{1}{\lambda}$

$\lambda = \frac{1}{7.5}$ / min.

$N = N_0 e^{-\lambda t} = N_0\ e^{-\left(\frac{1}{7.5}\right)\times 30} = N_0 e^{-4}$

$N = \frac{N_0}{e^4}$

8. $\frac{\sin i}{\sin 90}=\frac{1}{\sqrt{3.25}}$

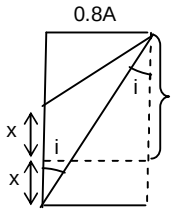
$\sin i = \dfrac{1}{\sqrt{3.25}}$

$= \dfrac{0.8A}{\sqrt{(A+x)^2+(0.8A)^2}} = \dfrac{1}{\sqrt{3.25}}$

$\therefore \quad \dfrac{0.64A^2}{(A+x)^2+(0.8A)^2} = \dfrac{1}{3.25}$

$x^2 + 2Ax - 0.44\,A^2 = 0$

$x = 0.2\,A$

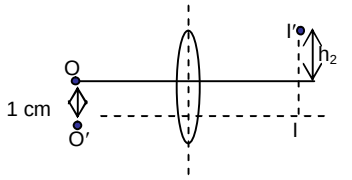


9. $\dfrac{1}{v} - \dfrac{1}{-15} = \dfrac{1}{10}$

$\Rightarrow$ v = 30 cm.

$m = \dfrac{v}{u} = -2 = \dfrac{h_2}{-1}$

Shift of image = II′ = (2 + 1) = 3 cm.

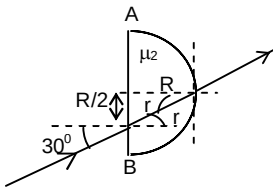


10. $\mu_1 \sin 30 = \mu_2 \sin r$

$\mu_1 = 1$ and $\tan r = \dfrac{1}{2}$

$\Rightarrow \sin r = \dfrac{1}{\sqrt{5}}$

$\Rightarrow \mu_2 = \dfrac{1/2}{1/\sqrt{5}} = \dfrac{\sqrt{5}}{2}$



11. $\dfrac{1}{u} + \dfrac{1}{v} = \dfrac{2}{R}$

$\Rightarrow \dfrac{1}{-15} + \dfrac{1}{v} = -\dfrac{1}{10}$

$\Rightarrow$ v = –30 cm.

$v_I = \left|\dfrac{dh_2}{dt}\right| = \left|-\dfrac{v}{u}\dfrac{dh_1}{dt}\right| = \left|-\dfrac{v}{u}.v_0\right| = \left|\dfrac{30}{15}\times 2\right| = 4\text{mm/s}.$

12. Resultant intensity is

$I_R = 4I\cos^2 \phi/2$

$I_{max} = I_0 = 4I$

$I_R = I_0 \cos^2(\pi/6) = \dfrac{3I_0}{4}$

13. Number of lines obtained $= \dfrac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$ $\quad n_2 = 7$

$= \dfrac{(7-2)(5+1)}{2} = 15$ $\quad n_1 = 2$

14. Number of nuclei of z will increase till the rates will become equal

i.e. $k = \lambda N_{max}$ or $k - \lambda N = \frac{dN}{dt}$,

$\Rightarrow \quad N_{max} = \frac{k}{\lambda}$

15. Maximum K.E. of the photoelectron is

$KE_{max} = \frac{hc}{\lambda} - \phi$

$= \frac{1242 eV - nm}{280 nm} - 2.5 eV = 1.9$ eV

since $0 \le$ k.E. ≤ 1.9 eV

$\therefore$ K.E. $\neq 2.4$ eV.

16. ${}_zX^A \rightarrow {}_{z-2}X^{A-8} + n\,{}_2He^4 + m_{-1}\beta^0$

$4n = 8$

$2n - m = 2$

$n = 2, m = 2$

17. Angular fringe width

$= \Delta\theta = \frac{\lambda}{d}$

If $\lambda \rightarrow 2\lambda$ & $d \rightarrow d/2$

then new fringe width $= \frac{4\lambda}{d} = 4\Delta\theta$

18. $\frac{hc}{\lambda} = z^2 Rhc\left(\frac{1}{(n-1)^2} - \frac{1}{n^2}\right)$

$\Rightarrow \quad \lambda \propto n^3$

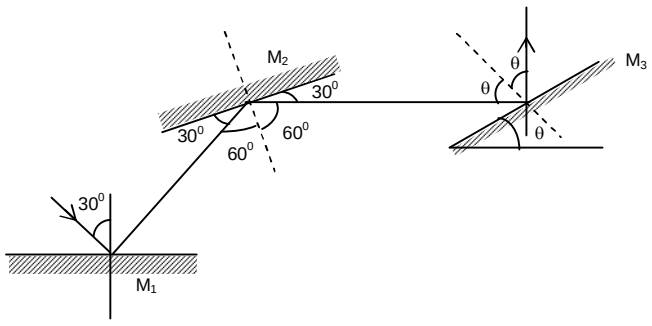
20. $\frac{\Delta A}{A} = \frac{\Delta \ell}{\ell} + \frac{\Delta b}{b}$

$\Delta A = \left(\frac{0.1}{10.0} + \frac{0.01}{1.00}\right) \times (10.0 \times 1.00) cm^2 = 0.2\ cm^2$

21. $\theta + \theta = 90^0$

$2\theta = 90^0$

$\theta = 45^0$



25. Energy received at photo sensitive surface per unit time $= \frac{ks}{4\pi a^2}$

Number of photons = $\dfrac{\frac{ks}{4\pi a^2}}{\frac{hc}{\lambda}} = \dfrac{k\lambda s}{4\pi hca^2}$

Maximum energy = $\left(\dfrac{hc - \lambda\phi}{\lambda}\right)$

26. $F = -\dfrac{dU}{dr} = \dfrac{he^2}{2r^2}$ . . . (i)

$F = \dfrac{mv^2}{r} = \dfrac{he^2}{2r^2}$ . . . (ii)

Also $mvr = \dfrac{h}{2\pi}$ . . . (iii)

$\therefore$ solving equation we get $r = \dfrac{h}{2\pi^2 me^2}$.

Chemistry

PART – II

1. Total pressure of a mixture of 2 volatile liquids is given by

$P_{Total} = P_A^\circ X_A + P_B^\circ X_B = \left(100 \times \frac{1}{4}\right) + \left(80 \times \frac{3}{4}\right)$ = (25 + 60) = 85 mm Hg

Mole fraction of A in the vapour phase (X'_A) is given by

$$X'_A = \frac{P_A}{P_T} = \frac{\left(100 \times \frac{1}{4}\right)}{85} = \frac{25}{85} = 0.294$$

$\therefore \quad X'_B = 0.706$

These mole fractions will go into the distillate. The total vapour pressure of the distillate would be

$P'_{Total} = P_A^\circ X'_A + P_B^\circ X'_B$ = (100 × 0.294) + (80 × 0.706)

= (29.4 + 56.48) = **85.88 mm Hg**.

2. At the same temperature, isotonic solutions will have same concentrations.

$Na_2SO_4 \rightleftharpoons 2Na^+ + SO_4^{2-}$

$1 - \alpha \qquad 2\alpha \qquad \alpha$

$C_{Na_2SO_4} = C_{glucose}$

$0.04(1 + 2\alpha) = 0.1$

α = **0.75**

3. $2KI + HgI_2 \longrightarrow K_2[HgI_4] \rightleftharpoons 2K^+ + [HgI_4]^{2-}$

Thus, there is net decrease in number of ions present in solution and freezing point is raised

4. $\Delta T_f = i \times K_f \times m$

$0.0054 = i \times 1.80 \times 0.001$

$i = 3$

$i = 1 + (n - 1)\alpha$

$n = 3$

$\therefore$ The correct formula of the compound is $[Pt(NH_3)_4Cl_2]Cl_2$

5. $As_2S_3 + (NH_4)_2S_2 \longrightarrow \underset{\text{soluble}}{(NH_4)_3AsO_3}$

9. The number of units of AB in a unit cell (fcc) = 4

Now, $\frac{r_+}{r_-}$ for octahedral co-ordination = 0.414

$\therefore r_+ = 100 \times 10^{-10} = 10^{-10}$ cm

$\frac{r_+}{r_-} = 10^{-10}$

So, $r_- = 241.5$ pm

Hence, (A).

10. $E = E^o - \frac{0.0591}{n}\log\frac{(Cr^{3+})}{(Cr_2O_7^{2-})^{1/2}(H^+)^7}$

11. $Pt \mid H_2 \mid H^+ \parallel H^+ \mid H_2 \mid Pt$

$1\,atm \quad x \quad 10^{-4}\,1\,atm$

$E_{cell} = \frac{0.0591}{2}\log\frac{(10^{-4})^2}{x^2}$

$x = 10^{-6}\,M$

13. $Ag^+ + e^- \longrightarrow Ag\,;\ E^o = 0.799\ V;\ \Delta G^o = -7.71\times10^4\ J$

$Ag(NH_3)_2^+ \longrightarrow Ag^+ + 2NH_3\,;\ K = 6\times10^{-8};\ \Delta G^o = (-2.303\,RT)\log K = 4.12\times10^4\ J$

$\Delta G^o_{net} = (-7.71\times10^4\ J) + (4.12\times10^4\ J) = -3.59\times10^4\ J$

$E^o = \frac{-\Delta G^o}{nF} = 0.372\,V$

14. The total charge of Fe^{2+} and Fe^{3+} should be 2.
Let, the percentage of Fe^{3+} = x
Hence, $3 \times x + 2(0.93 - x) = 2$
$\therefore x = 0.14$

$\%\ of\ Fe^{3+} = \frac{0.14}{0.93}\times100 = 15\%$

17. It does not dimerise due to delocalization of odd electrons.

19. $2HBr + SO_3 \longrightarrow H_2O + Br_2 + SO_2;$

20. $Ti + 2I_2 \xrightarrow{500K}$ TiI_4 (Volatile stable compound) $\xrightarrow{1700K}$ $Ti + 2I_2$ (Pure metal)

21. Carnallite is an important ore of magnesium. It's formula is $KCl.MgCl_2.6H_2O$.

23. $PbS + 2PbO \longrightarrow 2Pb + SO_2,\ SnO_2 + C \longrightarrow Sn + CO,\ SnO_2 + 2CO \longrightarrow Sn + 2CO_2.$

25. $VOSO_4 \Rightarrow VO^{2+} + SO_4^{2-}$
So oxidation state of V is +4.

26. $CO_2 + H_2O \rightleftharpoons H_2CO_3$

$CrO_4^{2-} + 2H^+ \rightleftharpoons \underset{(orange\ red)}{Cr_2O_7^{2-}} + H_2O$

27. $Mn^{2+}: 3d^5 4s^0$
In presence of weak field ligand, three will be no pairing of electrons, so it will form a high spin complex i.e. the number of unpaired electrons = 5.

Mathematics PART – III

SECTION – A

1. Let $\vec{c} = (c_1, c_2, c_3)$, then

$|\vec{c}| = |\vec{a}| = |\vec{b}| = \sqrt{2} = \sqrt{c_1^2 + c_2^2 + c_3^2}$

It is given that the angles between the vectors are identical and equal to φ (say), then

$\cos\varphi = \dfrac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|} = \dfrac{0+1+0}{\sqrt{2}\sqrt{2}} = \dfrac{1}{2}$

$\dfrac{\vec{a}.\vec{c}}{|\vec{a}||\vec{c}|} = \dfrac{c_1 + c_2}{2} = \dfrac{1}{2}$ and $\dfrac{\vec{b}.\vec{c}}{|\vec{b}||\vec{c}|} = \dfrac{c_2 + c_3}{2} = \dfrac{1}{2}$

Hence $c_1 + c_2 = 1, c_2 + c_3 = 1$ and $c_1^2 + c_2^2 + c_3^2 = 2$

On solving the equation, we get $c_1 = 1, -\dfrac{1}{3}; c_2 = 0, \dfrac{4}{3}; c_3 = 1, -\dfrac{1}{3}$

Hence co-ordinates of c are $(1, 0, 1)$ or $\left(-\dfrac{1}{3}, \dfrac{4}{3}, \dfrac{-1}{3}\right)$.

Trick : Obviously, length of the vector $(1, 0, 1)$ i.e., $\hat{i} + \hat{k}$ is equal to length of $\vec{a}$ and $\vec{b}$. Also it makes equal angle with $\vec{a}$ and $\vec{b}$ and equal to that of between $\vec{a}$ and $\vec{b}$ i.e. , $\dfrac{\pi}{3}$.

2. Since the vectors $\vec{a} = cx\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{b} = x\hat{i} + 2\hat{j} + 2cx\hat{k}$ make an obtuse angle, therefore

$\vec{a}.\vec{b} < 0 \Rightarrow cx^2 - 12 + 6cx < 0$ for all x

$\Rightarrow c < 0$ and Discriminant < 0

$\Rightarrow c < 0$ and $36c^2 + 48c < 0$

$\Rightarrow c < 0$ and $3c + 4 > 0 \Rightarrow c < 0$ and $3c + 4 > 0$

$\Rightarrow -\dfrac{4}{3} < c < 0.$

3. $[\vec{p}\,\vec{q}\,\vec{r}] = \vec{p}.(\vec{q} \times \vec{r})$

$= \dfrac{\vec{b} \times \vec{c}}{\vec{a}.\vec{b} \times \vec{c}}.\left[\dfrac{\vec{c} \times \vec{a}}{\vec{a}.\vec{b} \times \vec{c}} \times \dfrac{\vec{a} \times \vec{b}}{\vec{a}.\vec{b} \times \vec{c}}\right]$

$= \dfrac{(\vec{b} \times \vec{c}).\left[(\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b})\right]}{[\vec{a}.(\vec{b} \times \vec{c})]^3}$

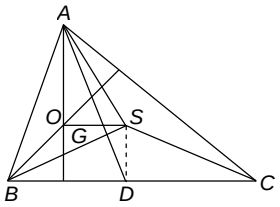
$\dfrac{(\vec{b} \times \vec{c}).\left[\{(\vec{c} \times \vec{a}).\vec{b}\}\vec{a} - \{(\vec{c} \times \vec{a}).\vec{a}\}\vec{b}\right]}{[\vec{a}.(\vec{b} \times \vec{c})]^3}$

$= \dfrac{(\vec{b} \times \vec{c}) \cdot \left[\{(\vec{c} \times \vec{a}).\vec{b}\}\vec{a}\right]}{\left[\vec{a}.(\vec{b} \times \vec{c})\right]^3}, \qquad \{\because (\vec{c} \times \vec{a}).\vec{a} = 0\}$

$= \dfrac{(\vec{b} \times \vec{c})\left[\{(\vec{a}.(\vec{b} \times \vec{c})\}\vec{a}\right]}{[\vec{a}.(\vec{b} \times \vec{c})]^3} = \dfrac{[\vec{a}.(\vec{b} \times \vec{c})]\,[(\vec{b} \times \vec{c}).\vec{a}]}{[\vec{a}.(\vec{b} \times \vec{c})]^3}$

$= \dfrac{1}{\vec{a}.\vec{b}\times\vec{c}}.$

4. Let P be any point in the plane of the triangle ABC. Then



$\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}=\overrightarrow{PA}+(\overrightarrow{PB}+\overrightarrow{PC})=\overrightarrow{PA}+2\overrightarrow{PD}$

$=(1+2)\overrightarrow{PG}=3\overrightarrow{PG}$

Since G divides AD in the ratio $2:1$.

$\therefore\ \overrightarrow{SA}+\overrightarrow{SB}+\overrightarrow{SC}=3\overrightarrow{SG}=2\overrightarrow{SG}+\overrightarrow{SG}$

$=\overrightarrow{GO}+\overrightarrow{SG}=\overrightarrow{SO}$, $(\because\ OG=2SG)$.

5. $[\vec{a}\vec{b}\vec{c}]\neq 0$

Also $|[\vec{a}\vec{b}\vec{c}]| = [\vec{b}\times\vec{c}\ \vec{c}\times\vec{a}\ \vec{a}\times\vec{b}] = [\vec{a}\vec{b}\vec{c}]^2$

or $t^2 = |t| \Rightarrow t^2 = \pm t$, t = 0, 1, -1

but $[\vec{a}\vec{b}\vec{c}]\neq 0$, $[\vec{a}\vec{b}\vec{c}] = 1$

and $[\vec{a}\vec{b}\vec{c}] = -1$

6. $\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)=\vec{a}\times\left(\vec{a}\times\vec{b}\right)+\vec{b}\times\left(\vec{a}\times\vec{b}\right)=\left(\vec{a}.\vec{b}\right)\vec{a}-|\vec{a}|^2\vec{b}+|\vec{b}|^2\vec{a}-\left(\vec{a}.\vec{b}\right)\vec{b}$

$=\vec{a}.\vec{b}\left(\vec{a}-\vec{b}\right)-\vec{b}+\vec{a}\qquad(|\vec{a}|=|\vec{b}|=1)$

$=\left(\left(\vec{a}.\vec{b}\right)-1\right)\left(\vec{a}-\vec{b}\right)$

Hence $\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)$ is parallel to $\vec{a}-\vec{b}$

7. Let $\dfrac{x+1}{3}=\dfrac{y+3}{5}=\dfrac{z+5}{7}=\lambda$ .....(i)

Then $x=3\lambda-1,\ y=5\lambda-3, z=7\lambda-5$

General point on this line is $(3\lambda-1,5\lambda-3,7\lambda-5)$

Again let $\dfrac{x-2}{1}=\dfrac{y-4}{3}=\dfrac{z-6}{5}=\mu$ .....(ii)

Then $x=\mu+2, y=3\mu+4, z=5\mu+6$

A general point on this line is $(\mu+2,3\mu+4,5\mu+6)$

For intersection, they have a common point, for some values of λ and μ, we must have $bcx+cay-abz=0$,

$(5\lambda-3)=(3\mu+4),\quad(7\lambda-5)=(5\mu+6)$

From first two we have, $\mu=3\lambda-3$ ……(iii)

and $3\mu=5\lambda-7$ .....(iv)

From (iii), put the values of μ in (iv), we have $3(3\lambda-3)=5\lambda-7$

$\Rightarrow 9\lambda - 9 = 5\lambda - 7$ or $4\lambda = 2$ or $\lambda = \frac{1}{2}$

Put $\lambda = \frac{1}{2}$ in (iii), we get $\mu = -\frac{3}{2}$ (Putting $\lambda = \frac{1}{2}$)

The required point of intersection is

$\left[\frac{3}{2} - 1\right], \left[\frac{5}{2} - 3\right], \left[\frac{7}{2} - 5\right] = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right]$.

8. $\theta = \sin^{-1}\left(\frac{6 + 2 - 12}{\sqrt{14}.\sqrt{29}}\right) \Rightarrow \theta = \sin^{-1}\left(\frac{-4}{\sqrt{406}}\right)$.

9. The equation of plane, in which the line $\frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5}$ lies, is

$A(x-5) + B(y-7) + C(z+3) = 0$ ……(i)

Where $4A + 4B - 5C = 0$ ……(ii)

Also, since line $\frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$ lies in this plane

$\therefore\ 7A + B + 3C = 0$ ……(iii)

By (ii) and (iii), we get $\frac{A}{17} = \frac{B}{-47} = \frac{C}{-24}$

$\therefore$ The required plane is

$17(x-5) - 47(y-7) + (-24)(z+3) = 0$

$\Rightarrow\ 17x - 47y - 24z + 172 = 0$.

10. Equation of a plane passing through origin is $ax + by + cz = 0$, but it also passes through (1, – 2, – 1) and (3,–2,3), therefore $a - 2b - c = 0$ and $3a - 2b + 3c = 0$.

$\Rightarrow a = -2c$ and $b = -\frac{3c}{2}$

$\therefore$ Plane is $-2cx - \frac{3c}{2}y + cz = 0 \Rightarrow 4x + 3y - 2z = 0$.

Hence direction cosines of plane are $\left(\frac{4}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{-2}{\sqrt{29}}\right)$

11. Let A ≡ (1, –1, 0), B ≡ (–1, 0, 1)

$\therefore$ direction ratios of segment AB are (2, –1, –1). If 'θ' be the acute angle between segment AB and normal to plane,

$\cos\theta = \frac{|2.2 - 1.1 - 1.6|}{\sqrt{4+1+36}.\sqrt{4+1+1}} = \frac{3}{\sqrt{246}}$

Length of projection = (AB) $\sin\theta = \sqrt{6}\ .\ \sqrt{1 - \frac{9}{246}} = \sqrt{\frac{237}{61}}$ units.

12. Given line passes through (1, 2, 4) and this point also lies on the given plane.

Thus, required line will be in the form of $\frac{x-1}{l} = \frac{y-2}{m} = \frac{z-4}{n}$.

Any point on the given line is $(-r_1 + 1, 3r_1 + 2, r_1 + 4)$.

If $r_1 = 1$, this point becomes P = (0, 5, 5)

Let Q = (a, b, (C) be the reflection of P in the given plane, then

$\frac{a}{2}.1+\frac{b+5}{2}.1+\frac{5+c}{2}.1=7$

i.e., a + b + c = 4, and $\frac{a}{1}=\frac{b-5}{1}=\frac{c-5}{1}=\lambda$ (say)

$\Rightarrow a=\lambda, b=5+\lambda, c=5+\lambda$

$\Rightarrow 10+3\lambda=4 \Rightarrow \lambda=-2$

Thus, $Q \equiv (-2, 3, 3)$

Hence, direction ratios of reflected line are (–3, 1, –1)

Thus, it's equation is $\frac{x-1}{-3}=\frac{y-2}{1}=\frac{z-4}{-1}$

13. In first trial he get's a head = ½

probability that say in first trial he get's a trial then in dice throwing we should get (1 to 4) and then head comes. Probability of that $=\frac{1}{2}\times\frac{4}{6}\times\frac{1}{2}=\frac{1}{6}$

So total probability will become $=\frac{1}{2}+\frac{1}{6}+\frac{1}{18}+...=\frac{1/2}{1-1/3}=\frac{3}{4}$

14. No one dies $\Rightarrow$ probability = $(1-p)^n$

atleast one dies $\Rightarrow 1 = 1-(1-p)^n$

now $A_1, A_2, \ldots A_n$ can make n! cases between them if the fix A_1 to die first then rest $A_2 \ldots A_n$ will make (n –1)! cases between them so probability of A_1 dies first $=\frac{(n-1)!}{n!}=\frac{1}{n}$

So, total probability of A_1 dies and dies first $\Rightarrow \frac{1}{n}\left(1-(1-p)^n\right)$.

15. We have ${}^{100}C_{50}\,p^{50}(1-p)^{50} = {}^{100}C_{51}(1-p)^{49}$

or $\frac{1-p}{p}=\frac{100!}{51!\,.\,49!}\times\frac{50!\,.\,50!}{100!}=\frac{50}{51}$

or $51-51p=50p$

$\Rightarrow p=\frac{51}{101}$.

16. The man will be one step away from the starting point if (i) either he is one step ahead or (ii) one step behind the starting point.

$\therefore$ The required probability $= P(i)+P(ii)$

The man will be one step ahead at the end of eleven steps if he moves six steps forward and five steps backward. The probability of this event $= {}^{11}C_6(0.4)^6(0.6)^5$.

The man will be one step behind at the end of eleven steps if he moves six steps backward and five steps forward.

The probability of this event $= {}^{11}C_6(0.6)^6(0.4)^5$.

Hence the required probability

$= {}^{11}C_6(0.4)^6(0.6)^5 + {}^{11}C_6(0.6)^6(0.4)^5$

$= {}^{11}C_6(0.4)^5(0.6)^5(0.4+0.6) = {}^{11}C_6(0.24)^5$.

17. $P(x>1.5)=1-P(x=0)-P(x=1)$ λ : expected minimum positive integer

$P(x = k) = e^{-\lambda}\frac{\lambda^k}{k!}$ e : Euler number.

$\therefore\ P(x > 1.5) = 1 - \frac{1}{e^2} - \frac{2}{e^2} = 1 - \frac{3}{e^2}$.

18. $\Delta = 1(\omega^{3n} - 1) + \omega^n(\omega^{2n} - \omega^{2n}) + \omega^{2n}(\omega^n - \omega^{4n})$

$\Delta = [(\omega^3)^n - 1] + 0 + \omega^{2n}[\omega^n - (\omega^3)^n.\omega^n]$

$\Delta = 1 - 1 + 0 + \omega^{2n}[\omega^n - \omega^n] = 0$.

19. As we have $A^2 = 2A - I \Rightarrow A^2.A = (2A - I)A$

$\Rightarrow A^3 = 2A^2 - IA = 2(2A - I) - A \Rightarrow A^3 = 3A - 2I \quad \{\because IA = A \text{ and } A^2 = 2A - I\}$

Similarly, $A^4 = 4A - 3I, A^5 = 5A - 4I$..

Hence $y^b = e^m, x^c y^d = e^n$.

20. We have $a_1, a_2, a_3 ..... a_n$ in G.P.

then $r = \frac{a_2}{a_1}$ i.e., $r = \frac{a_{n+1}}{a_n} = \frac{a_{n+2}}{a_{n+1}} = ...$

Hence $\log_r = \log(a_{n+1}) - \log(a_n) = \log(a_{n+2}) - \log(a_{n+1}) = ...$

Now $\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$

Operate $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_2$

$= \begin{vmatrix} \log a_n & (\log a_{n+1} - \log a_n) & (\log a_{n+2} - \log a_{n+1}) \\ \log a_{n+3} & (\log a_{n+4} - \log a_{n+3}) & (\log a_{n+5} - \log a_{n+4}) \\ \log a_{n+6} & (\log a_{n+7} - \log a_{n+6}) & (\log a_{n+8} - \log a_{n+7}) \end{vmatrix}$

$= \begin{vmatrix} \log a_n & \log r & \log r \\ \log a_{n+3} & \log r & \log r \\ \log a_{n+6} & \log r & \log r \end{vmatrix} = 0$.

21. $\Delta^c = \Delta^{n-1} = \Delta^{3-1} = \Delta^2 = (11)^2 = 121$.

But we have to find the value of the square of the determinant, so required value is $(121)^2 = 14641$.

22. Since $(AB)^{-1} = B^{-1}A^{-1}$.

23. For the given matrix to be singular, we must have, $\begin{vmatrix} 3 & -1+x & 2 \\ 3 & -1 & x+2 \\ x+3 & -1 & 2 \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} 3 & -1+x & 2 \\ 0 & -x & x \\ x & -x & 0 \end{vmatrix} = 0, \ [R_2 \to R_2 - R_1, R_3 \to R_3 - R_1]$

$$\Rightarrow \quad \begin{vmatrix} x+4 & -1+x & 2 \\ 0 & -x & x \\ 0 & -x & 0 \end{vmatrix} = 0\,,\ [C_1 \to C_1 + C_2 + C_3]$$

$\Rightarrow \quad (x+4)(0+x^2) = 0 \Rightarrow x = -4, 0$

Note that only $-4 \in [-4,-1]$.

24. $-\frac{\pi}{2} \le \sin^{-1} x \le \frac{\pi}{2}$

$-\frac{\pi}{2} \le \sin^{-1} y \le \frac{\pi}{2}$

$-\pi \le \sin^{-1} x + \sin^{-1} y \le \pi$

so for $\sin^{-1} x + \sin^{-1} y = -\pi$

$\Rightarrow \sin^{-1} x = -\frac{\pi}{2}$ & $\sin^{-1} y = -\frac{\pi}{2}$

$x = -1$ and $y = -1$

area required $1 - \left| \int_{-1}^{0} -(x+1)^2 dx \right| = 1 - \frac{1}{3} = \frac{2}{3}$ sq. units.

25. The curves given by $y = x + \sin x$ and $y = f^{-1}(x)$ are images of each other in the line $y = x$.

Hence required area = $\int_0^{\pi} \big((x + \sin x) - x\big) dx = -\big[\cos x\big]_0^{\pi} = 2$

26. Circle has (2, –1) as its centre and radius of this circle is 2.

Thus if P(x, y) be any point on it, then $x \in [0, 4]$.

Let $g(x) = \frac{x^3}{100} + \frac{x}{50}$

$g'(x) = \frac{3x^2}{100} + \frac{1}{50} > 0 \ \forall x \in (0,4]$

The g(x) is increasing in [0, 4]

$g(0) = 0, g(4) = \frac{18}{25}$

$g(x) = \left[0, \frac{18}{25}\right] \forall x \in [0,4] \Rightarrow [g(x)] = 0 \ \forall \ x \in [0, 4]$

$y = \left[\frac{x^3}{100} + \frac{x}{50}\right]$, represents the x-axis.

CA = CB = 2, CD = 1

$\cos\theta = \frac{CD}{CA} = \frac{1}{2} \ \theta = \frac{\pi}{3}$

$\angle ACB = \frac{2\pi}{3}, \Delta ACB = \frac{1}{2} \times 2^2 \times \sin\frac{2\pi}{3} = \sqrt{3}$

Area of sector ACB $= \frac{1}{2} \times 2^2 \times \frac{2\pi}{3} = \frac{4\pi}{3}$ sq. units

27. Since $\int_0^a f(x)dx = \frac{a^2}{2} + \frac{a}{2}\sin a + \frac{\pi}{2}\cos a$

Differentiating both sides w.r.t. a, we get

$\Rightarrow \quad f(a) = a + \frac{a}{2}\cos a + \frac{1}{2}\sin a - \frac{\pi}{2}\sin a$

$\therefore \quad f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + 0 + \frac{1}{2}.1 - \frac{\pi}{2}.1 = \frac{1}{2}$

28. For $x \in [1, 4/3]$, $y = \sqrt{\sin 1 + 0} = \sqrt{\sin 1}$

So required area $= \int_1^{4/3} \sqrt{\sin 1}\, dx = \frac{1}{3}\sqrt{\sin 1}$ Sq. units.

29. $f(x) = \frac{x^2}{6}$ $A = \frac{1}{6}\int_0^3 x^2 dx = 3/2$

30. $y = (c_1 + c_2)\cos(x + c_3) - c_4 e^{x+c_5}$

$y_1 = -(c_1 + c_2)\sin(x + c_3) - c_4 e^{x+c_5}$

$y_2 = -(c_1 + c_2)\cos(x + c_3) - c_4 e^{x+C_5} = -y - 2c_4 e^{x+c_5}$

$\Rightarrow y_3 = -y_1 - 2c_4 e^{x+c_5} = -y_1 + y_2 + y$

Hence the differential equation is $y_3 - y_2 + y_1 - y = 0$, which is of order 3.